ROBODOR

KR380 AC MOTOR CONTROL UNIT FOR HIGH SPEED ROLLING DOOR AND SECTIONEL DOOR SYSTEMS



CE

KR380 V1.0-S1

2020

Important!
You must read the job description before operating, connecting and starting the door control system!

CONTENT

1. SAFETY INFORMATION 3
2. TECHNICAL INFORMATION 4
3. INSTALLING THE CONTROL UNIT 5
4. ELECTRICAL CONNECTION 6
5. MOTOR CONNECTION 6
6. KEYBOARD GENERAL OPERATION 6
6. MENU 6
6.1. COUNTER 7
6.2. OPERATION MODE SETTINGS 7
6.3. LEARNING MODE 8
6.4. LIMIT POSITION FINE TUNINING 8
6.4.1. OPEN POSITION FINE TUNINING 9
6.4.2. CLOSE POSITION FINE TUNINING 9
6.5. RAMPING ADJUSTMENT 9
6.6. SPEED ADJUSTMENT 9
6.7. TORQUE ADJUSTMENT 10
6.8. MOTOR INFORMATION 10
6.9. SCREEN ADJUSTMENT 10
6.10. WAITING TIME (Sn) 11
6.11. LANGUAGE (ENGLISH / TURKISH) 11
6.12. MOTOR DIRECTION 11
6.13. SYSTEM SETTINGS 12
6.14. RELAY SETTINGS 14
6.15. SYSTEM RESET 17
7. OTHER CONNECTIONS 17
7.1. Brake Circuit Connection 17
7.2. Emergency Button 17
7.3. Digital Limit Switch 17
7.4. Mechanical limit switch connection 18
7.5. Photocell 18
7.6. OSE 18
8. CONTROL UNIT SOCKET ENTRIES 20
9. TERMS OF WARRANTY 21

1. SAFETY INFORMATION

⚠ ATTENTION	COMPLIANCE WITH THESE INSTRUCTIONS IS IMPORTANT FOR PERSONAL SAFETY. FOLLOWING INSTRUCTIONS: Failure to comply with the safety information can be hazardous to health and damage the control system.

• Installation, operation and maintenance should only be carried out by qualified personnel.

• The control system should only be opened when the power supply is completely switched off.

• Potential free-contacts or other contact points of the redundant outputs are still working after the control system has been switched off or when the main plug is not plugged in or operating with main external power or dangerous voltage. The appropriate warning label must be clearly visible on the protective system packaging.

• The control system must never be switched on.

• If the connection cable of the appliance is damaged, it must be replaced by the manufacturer or its after-sales service or a qualified person to avoid danger.

• There are some voltages on the DC capacitor for up to 1/2 minute even after switching off the power supply. The discharge time for the voltage values to fall below 1 V DC is maximum 1 minute. It is dangerous to touch the internal control system components during discharge.

• If the 24 V control system is short-circuited or overloaded, the tripping power supply is not switched on even when the DC capacitors are charged.

• It is not possible to turn on the power supply unit without eliminating the short circuit or overload problem.

• After the power is turned off, the power supply is powered by the DC capacitor for a few seconds and the power supply continues for a certain time depending on the load of the power supply unit.

• The control system must not be operated without a connected protective earth. The absence of protection grounding can cause dangerous voltages due to the discharge capacity of the control system housing.

• The system must not be opened or operated while it is wet. This may cause damage to the control system.

• If the control system is to be used outside the specified temperature range, it must be used in the set and supervised heating system so that the specified operating temperature can be maintained when the power supply is turned on and during the control system operation.

• The control system must not be operated if the keypad or display is damaged. Damaged keypads and displays should be replaced. Avoid touching the keypad with sharp objects to avoid damaging the keypad. Only touch the keypad with fingers.

• Before starting up the control system for the first time, ensure that the plug connections and evaluation cards (portable modules) are in the correct position. Improperly installed connections can damage the control system, such as installing unauthorized cards.

• If the door is operating in manual mode, it must be ensured that the door area is visible to the operator, because in this manual mode safety devices such as safety bars and light barriers are not active.

• Check parameter settings and safety equipment operation. The parameters, bridges and other operating elements must be set by qualified personnel.

2. TECHNICAL INFORMATION

Dimensions (GxYxD):	230*330*130 mm (Except for Emergency STOP)
External Power Supply	380 VAC ± 10%, 50...60 Hz Input
Fuses	1 x 20A K-Type ve 1x1A K-Type
Controller Inputs	24 VDC 15 mA, max. 26VDC / 20mA < 5 V: passive logic 0; > 7 V: aktive > logic 1 Minimum signal duration for inputs > 100 ms, Opto coupler at the inputs Galvanic isolation
Power Consuption	Max. 8W (Power supply without motor)
Output Power	380VAC, 2200W Maximum length of motor cable 10m.
Internal Power Supply	24V DC 650 mA 16W SMPS
RS485 A ve B	RS485 level for digital limit switch only, terminates 120Ω
Security Chain / E-STOP	Contact load capacity: ≥ 26 VDC / ≥ 120 mA The system will stop working if the security chain is interrupted.
Electrical Safety Edge	The Electrical Safety Edge is terminated with 1.2kΩ or 8.2kΩ. Rated value 11.5V / maximum 130mA
Electrical Regulators Outputs	If the loads are switched on (other electrical regulators or brakes), they must be equipped with the corresponding noise canceling devices (free-wheeling diodes, varistors, RC elements).
Brake Relay	220VAC The brake coil is driven. Energized during door movement.
Relays: Relay 1, Relay 2, Relay 3:	5A Relays with NO / NC Output. Programmable for Error / Door Position Messages / Lamp Functions.
Brake Resistor (Optional):	Maximum 50W, 0.5 seconds, repeat rate 20 sec per minute.
Operating temperature range	-10…+50C (working), -20…+70C (storage)
Humidity in the air	Up to 95%
Vibration	Low vibration setup, eg. on brick wall.
IP Class	IP54 (IP65 replacing CEE connection and stamping cable entry) All unused cable entries must be closed.
Weight - Mounting	Approx. 3150 gr - Vertical to the Wall
Main cable installation	Installation of the main cable must only be carried out by the manufacturer, after-sales service or other qualified personnel.
Material Type	Motor equipment, external motor is not included.

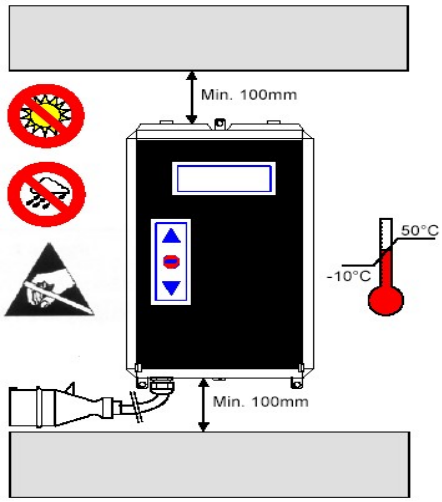
We do not claim that the security information is fully covered. Please contact the service for more information about the product.

3. INSTALLING THE CONTROL UNIT

CAUTION: IMPORTANT INFORMATION FOR SECURITY INSTALLATION.
FOLLOW ALL INSTRUCTIONS, MISUSE INSTALLATION MAY CAUSE SERIOUS INJURY.

WARNING During the installation of the control system, the system must not be connected to the mains.

Prior to installation, the control unit must be checked to ensure that it is not damaged by transportation or any other damage.



Damage within the control system leads to significant indirect damage to the control system and the risk of injury to the user.
It is forbidden to touch electronic components, especially the components of the processor current. Electronic components can be damaged or destroyed by electrostatic discharge.

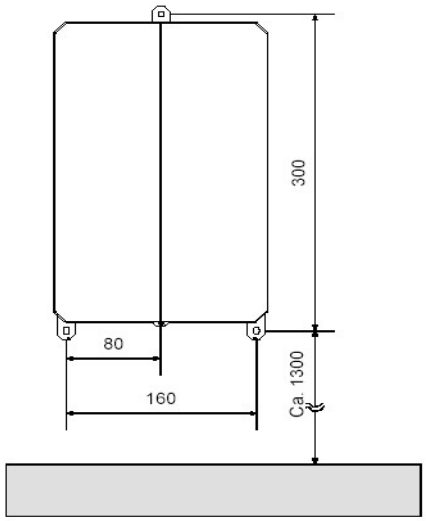
Before opening the cover lid, make sure that there are no drilling fillings on the cover that could fall into the cover.

The system must not be connected to the mains during installation.

Unused cable entries must be sealed using suitable tools to ensure the IP54 protection class of the coating.

The cable entries must not be subjected to any mechanical loading, in particular voltage loading.

Figure 1: Setting up the control system



Connection, testing and maintenance on the open control system must be carried out without voltage connection. The safety information must be observed.

There is a dangerous voltage of up to 2 minutes after the control system is switched off.

Touching electronic components is dangerous due to residual voltage.

The control system must not be operated with the housing cover open.

Figure 2: Drill panel Electrical Connection

ATTENTION When the system is turned on for the first time after the electrical installation is complete, it must be checked that all motor connections on the control and the motor are tightened and that the motor is turned on correctly.

We recommend at least one additional insulation for all components to be connected to the control system at nominal voltage.

Very high electrostatic charges are used especially in fast-running plastic doors. This voltage discharge can damage the control system. Therefore, appropriate countermeasures must be taken to prevent electrostatic charging.

4. ELECTRICAL CONNECTION

KR380 control unit; externally 380VAC, 50Hz three-phase network is fed through. The three-phase and neutral input supply cables are taken from the panel. It is compatible with 380Vac, 50 / 60Hz mains frequency. The three-phase, neutral and ground terminals must be properly connected to the KR380 supply power inputs. The mains plug must be visible and accessible from the control system. The DC supply used by the Control Card is activated via an external SMPS via a socket on the card.

5. MOTOR CONNECTION

The control unit is designed for 380V asynchronous motors with electronic position encoders or limit limit switches. Motor connection is made by connecting the motor cable ends to the appropriate socket. The phase sequence of the motor leads is important. If reverse connection is made, direction of motor movement will change. During the initial technical service installation, the current connection direction must not be changed from the inside. The motor cable should have a minimum capacity of 2.5 mm^2 shielded cable to carry 20A.

If the Control Unit is used in PVC systems, it should be remembered that the motor magnetic relay terminals are connected to the brake relay.

6. KEYBOARD GENERAL OPERATION

There are three buttons on the keypad. They performed different tasks in different menus. In general, **UP**, **DOWN** and **STOP** buttons provide control of the KR380.



It is used to stop the motor movement if the **STOP** Button (short) is pressed in automatic mode, and to switch to the settings menu if (long) is pressed. If the **STOP** button is pressed (short) in the menus, the sub-menus are entered for the respective menu, the (long) presses allow you to move to the upper menus and save the settings. The **STOP** (long) button is used to change from the submenus to the upper menu.

In automatic mode and manual mode, the **UP** and **DOWN** buttons are used to start the motor movements in accordance with their respective directions.

Within the menus, it is used for increment and decrement in transition and value assignment between menus.

6. MENU

In the automatic mode screen, the **STOP** and **UP** buttons (long) are pressed together to switch to the setup menus. The PASSWORD screen is displayed before switching to the Settings menu.



The user increases and decreases the number of digits using the **UP** and **DOWN** keys, and when he reaches the number of digits he wants to confirm, he changes to the other digit field by confirming with the **STOP** key.

Similarly, after confirming the four-digit setting, if the password is correct, the confirmation text is displayed on the screen and the menus are accessed. Password is the number of **2013**.

If the **STOP** button (long) is pressed while the **PASSWORD** is in the interrogation screen or in the menus section, the automatic mode screen is returned.

6.1. COUNTER

Displays the number of times the door has been opened and closed in automatic mode operation since installation. Pressing the **STOP** button (long) returns to the main menu screen.



Shows the number of door openings since the first installation of the system.

6.2. OPERATION MODE SETTINGS



AUTOMATIC
MANUAL
MANUAL UNLIMITED
TEST

This is the menu where the operation mode of the door is determined. Normally the door is set to operate between the upper and lower limits in automatic mode. For the door system, four different modes are selected from the Operating Mode Setting menu: Automatic / Manual / Manual Limit / Test. With the **UP / DOWN** buttons, one of these modes is determined by pressing and holding the **STOP** button and switches to that mode. The mode of door operation is changed.



In the AUTOMATIC mode, the automatic opening and closing operation is enabled. At the same time, the automatic closing time of the door can be set in this mode.



In **MANUAL** mode, the movement of the door between the defined limits up and down can be controlled manually. **UP** and **DOWN** buttons are pressed to realize the number of doors in or out.



In MANUAL UNLIMITED mode, the movement of the gate can be moved by holding down the **UP** and **DOWN** buttons without limiting the movement of the gate to position limits. In **TEST** mode, the door is opened and closed continuously with the waiting times (sec) set in the Wait Time menu.



In test mode, the door automatically descends and exits according to the automatic closing time set. When **STOP** button is pressed, test automatic descent function stops.

6.3. LEARNING MODE





The Door Learning Menu is the section that stores the lower and upper limit values of the movement that the door will perform automatically.

In case of using electronic encoder when the panel is energized, the initial values UPPER LIMIT and DOWN LIMIT are randomly assigned. These initial values must be changed in the new door installation. To do this, press the **STOP** and **UP** buttons at the same time (long) to go to the settings menu and find the Door Learning Setting menu by using the up / down buttons. By pressing the **STOP** button (short) while this menu is on the display, the positioning system in the menu is entered.

In this menu, the upper and lower limits of the door are set during the initial installation phase. When this menu is entered, the system first wants the door to be moved to the lower position.

In learning mode, the controlled movement of the door is performed by holding down the **UP** or **DOWN** key. The door movement is first performed to determine the lower limit point of the closing point of the motor below.

This limit is saved by long pressing the **STOP** key when the door is moved to the desired lower position by pressing and holding the **UP** and **DOWN** buttons. Then the system automatically asks for the upper limit.



The position of the door brought to the lower limit point is stored in the memory by pressing the **STOP** key (long). As a result, DOWN COMPLETED appears on the second line of the display.



At this stage, the display asks the UP LIMIT point on the first line. After this step, the **UP** / **DOWN** keys are pressed again to move the door to the upper limit point where it is desired to stop.

When the desired upper limit point is reached, the **STOP** position (long) is pressed and the position of the door is displayed as LEARN COMPLETED and it is stored as upper limit and the mode is changed to automatic mode.

In this transition phase, since the door is finally at the upper limit point, the door starts to move automatically to the lower limit point by pressing the **DOWN** key (short). In automatic mode operation, the range of motion will be between the lower and upper limits.

Again when the desired upper limit is reached by pressing the **UP** and **DOWN** keys, it is saved at the upper limit by long pressing the STOP button. The system automatically switches to the main menu after recording is complete.

6.4. LIMIT POSITION FINE TUNINING

This menu is used if you want to set the upper and lower limits more accurately. Within the menu, there are two options, Upper Limit Fine Adjustment and Lower Limit Fine Adjustment.

The values entered for fine tuning can be changed between +100 and -100 units. Positive values lower the limits and negative values lower the limits.

6.4.1. OPEN POSITION FINE TUNINING

If precise changes are made to the upper position trained during the **LEARNING MODE**, the upper position value can be changed precisely by pressing the **UP / DOWN** buttons (short-short). The value is saved by a long press on **STOP**.

6.4.2. CLOSE POSITION FINE TUNINING

If precise changes are made to the sub-position trained during **LEARNING MODE**, the upper position value can be precisely changed to the **UP** / **DOWN** buttons (short-short) from this menu. The value is saved by a long press on **STOP** and returns to the Main menus.



This menu is used if you want to set the upper and lower limits more accurately. Within the menu, there are two options, Upper Limit Fine Adjustment and Lower Limit Fine Adjustment.





The values entered for fine tuning can be changed between +100 and -100. Positive values lower the limits and negative values lower the limits.





6.5. RAMPING ADJUSTMENT



Ramp adjustment settings is not active on the KR380 contol unit.

6.6. SPEED ADJUSTMENT



Speed adjustment settings is not active on the KR380 contol unit.

6.7. TORQUE ADJUSTMENT



The torque adjustment setting is not active on the KR380 control unit.

6.8. MOTOR INFORMATION



The Motor Information Input setting is not active on the KR380 controller.

6.9. SCREEN ADJUSTMENT



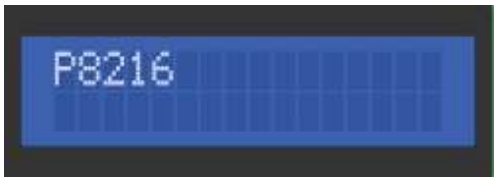
The system display can be adjusted to three different situations. Normally, the display shows verbal information about the operating mode of the door.





The encoder data is displayed on the test screen. This screen is available for service.





Test2 screen shows motor currents, line voltage, encoder data, load resistance current, motor frequency. This screen is available for service.







6.10. WAITING TIME (Sn)



Waiting menu is the place where the door stays above when the door is opened in automatic mode. The minimum is 2 seconds. To turn off the time, the value must be reduced until it says OFF. At the same time, this menu adjusts the door's up and down trigger time in test mode.





This value can be decreased by pressing the **UP** button (short-short) while in the Standby Time menu or by pressing the **DOWN** button (short-short). While the up and down keys serve as increments and decrements on this screen, pressing the **STOP** button (long) returns to the main menu.

6.11. LANGUAGE (ENGLISH / TURKISH)



Two different languages can be selected in this menu: Turkish and English. As the language in which the screen information is to be printed, one of the English / TURKISH language options is selected with the **UP** and **DOWN** buttons and the **STOP** button (long) is pressed to save the language and return to the main menu screen.





6.12. MOTOR DIRECTION



There are two different menus in this menu.





In the Motor Direction menu, the motor phases must be set in case of reversal, ie if the motor is reversed according to the buttons on the box. The standard version is “Normal”.

When the motor connection is made, the motor fixing direction is determined from this menu if the motor is to be fixed by turning it in the opposite direction and not by connecting the body as below. It is indicated that it is connected in the **REVERSE** direction when the UP button is pressed and in the **NORMAL** direction when the **DOWN** button is pressed. The value is saved by pressing the **STOP** button (long).

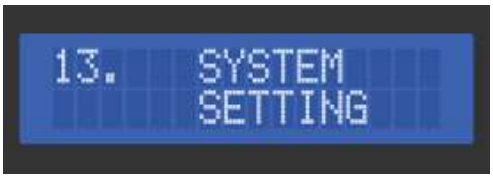


In the Encoder Direction menu, it is a setting that must be changed if the motor is installed in the opposite direction to the door. The standard version is set to “Normal.





6.13. SYSTEM SETTINGS



System settings can be made from this menu.



In the History Error List menu, the last error codes of the control unit can be accessed.



From the BOTTOM Security menu, the sensor type used for LOWER security can be selected as Photocell, OSE or off (Not used).



The use of the lower security system is disabled.



Photocell is chosen as the lower security system.

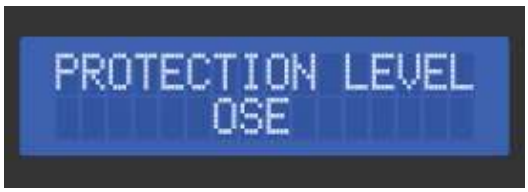


OSE Light Prison is chosen as the lower security system

Protection Level can be selected as Photocell or OSE from Security Level Setup menu.







If OSE SAFETY EDGE security system is required to be activated, OSE is activated by long-pressing the **STOP** button in this menu. If you want to turn it off, you can also set PASSIVE / OFF from the same menu and press the **STOP** button (long).



Limit Sensor Setting menu can be used to determine the type of sensor to be used for positioning. The model of the digital position encoder can also be selected. In addition to selecting Kostal, GFA brand electronic position encoders from the menu, it is also preferable to operate my system with mechanical limit switch.



Kostal brand digital encoder is selected as limit sensor



GFA brand digital encoder is selected as limit sensor.



Mechanical limit switch is preferred as limit sensor.

The activation position of the photocell and OSE safety guards connected in the system can be changed while the door is closing. By changing the number of positions, the position at which the sensors will be activated can be changed.

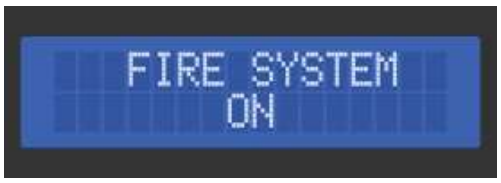


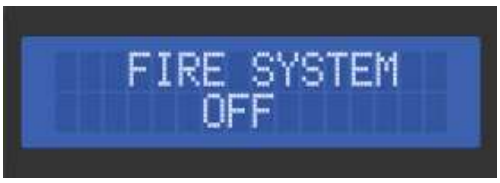


With the Fire Alarm System Setup menu, the Control Unit can activate the input from which the Fire Alarm Line is located within the building where it is installed.



When Fire Alarm On or Off is selected and recorded with **STOP** button, the Fire Alarm input of the card can be activated / deactivated.





6.14. RELAY SETTINGS



There are two programmable relays on the system. The operating status of these relays can be programmed from this menu.





In the Relay Settings menu, the settings of two programmable relays owned by the control card are made.

This menu assigns the tasks that each relay, including the Relay 1 and Relay 2 operating settings, must perform when the door is moving. Selecting Off disables the Relay. With the Normally Open selection, the relay door becomes the Normally Open Contact in the specified state, with the Normally Closed selection the relay door becomes the Normally Closed contact. If the Relay Flash / Relay Toogle operation mode is selected, the Relay repeats the on-off operation continuously if the On-Off time interval is desired.

RELAY OPERATION MODES

- OFF
- NORMALLY OPEN
- NORMALLY CLOSED

- RELAY TOGGLE



For relays; The mode setting has four modes, Passive, Normally Closed, Normally Open and Flash modes. By selecting the relay operation type in the relay setting menu, the system transfers the relay operation functions to the submenu for selection on the screen.



From here, the user determines the status of the door for the relay to operate. When the door is closed / door is opened / door is opened / door is closed / door is stopped / door is running / in case of error, one of the modes is selected and the job description is completed.

In the position setting, the position of the selected activity is selected via the mode setting of the relay.

RELAY ACTIVATION MODES

- DOOR CLOSED
- DOOR OPEN
- DOOR OPENING
- DOOR CLOSING
- DOOR STOPED

- DOOR MOVING
- DOOR ERROR

























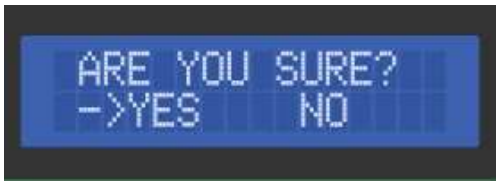




6.15. SYSTEM RESET



The system returns to the state where the first program was loaded with a special code entered from this menu. After this function, all setting information is restored to the factory settings and the door settings must not be reset.





When System Reset menu is entered, Yes and No options can be changed with Up and Down arrow keys. When YES is selected and confirmed with the **STOP** key, the system requests the Confirm password. The password is entered with the help of the buttons and the system is reset to the correct password.



7. OTHER CONNECTIONS

7.1. Brake Circuit Connection

The Brake Circuit is an electromechanical brake system connected to the motor shaft. It works with 220Vac supply. The board is fitted with a brake socket.

If the brake circuit terminals are not connected, the connection of the control card must be carefully checked for serious damage.

7.2. Emergency Button

When the emergency stop button on the control panel is pressed, the motor stops working. It cancels the power supply to the motor and brake.

7.3. Digital Limit Switch

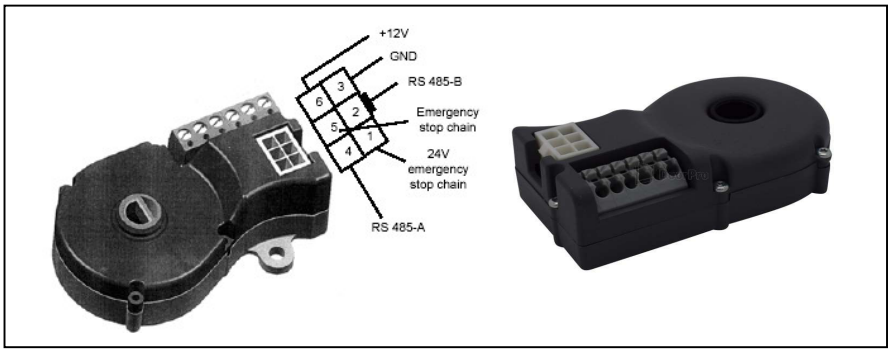


Figure 5 - Digital limit switch

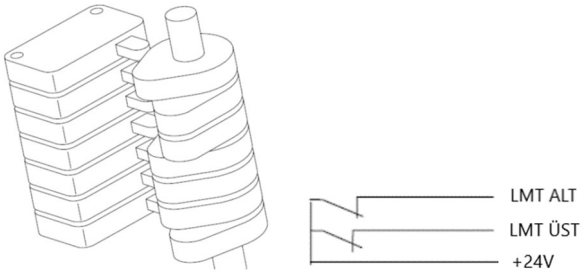
KR380 Digital limit switch system and mechanical limit switch can be used for positioning information in door control system.

The outputs of the digital limit switch fixed on the motor gear unit must be installed in the control cabinet during wiring. The cable harness must be connected from left to right at socket X10 6–4–2–3 during the delivery to +15 - A - B - GND terminals.

Connection Speed: 9600 bps, Connection Protocol: RS – 485

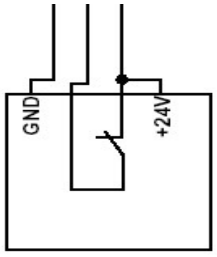
IMPORTANT	Check the electrical connection again before operating the control system. The device may be damaged by incorrect connections.
⚠ **WARNING**	To prevent unintended operation of the door, make the digital limit switch settings when the EMERGENCY STOP button is pressed or the control system is switched off.
IMPORTANT	If no safety system is connected, the door cannot be closed automatically.

7.4. Mechanical limit switch connection



If the position sensor to be used by selecting Limit Sensor sub-menu from System Setting menu is selected as Limit Switch switch, limit switch can be connected by using LMP UP, LMT DOWN and COM terminals in X6 socket.

7.5. Photocell



The light barrier system consists of the receiver and transmitter unit and the supply inputs are provided via the card. Only the receiver's signal is received on the control panel. The light barrier is only effective when the door is moved down. When it encounters an object, it is allowed to switch to manual mode by stopping the downward movement.

Figure 6: Photocell Connection

7.6. OSE



Safety edges are used wherever moving edges pose a hazard to people. The hazardous areas are protected by hollow rubber profiles. As soon as a person or object touches them, the sensing profile is deformed and the potentially hazardous movement stops.

Figure 7: OSE

KR380 Control Unit Test Standards Compliance

2014/30/EU EN 61000-6-1, EN 61000-6-3, 2014/35/EU EN 60335-1, 2006/42/EC EN ISO 12100:2010

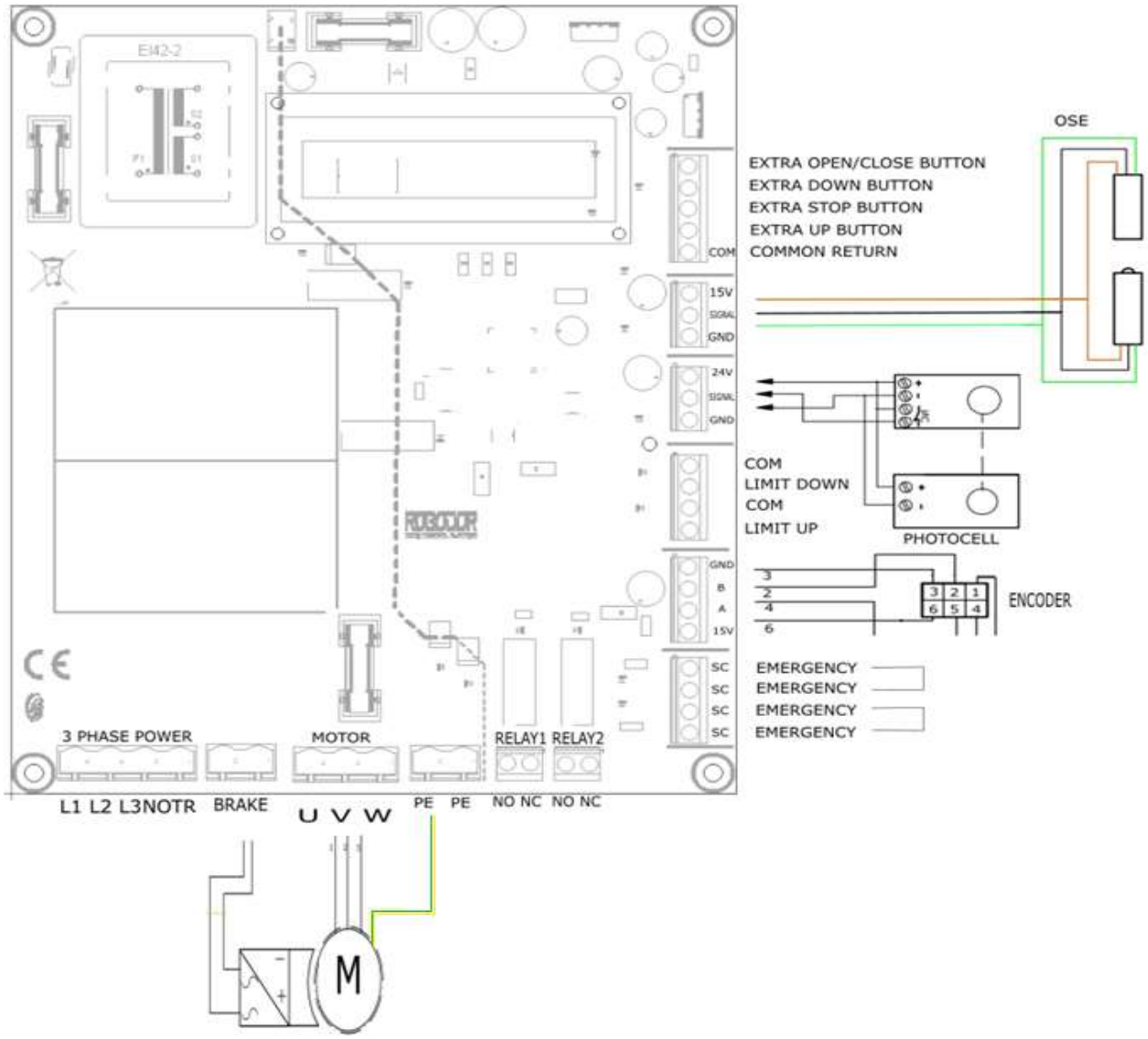





APPLIED STANDART / *UYGULANAN STANDART*	**APPLIED EXPERIMENTS / *UYGULANAN DENEYLER***	**RESULT / *SONUÇ***
EN 61000-6-1:2007 EN 61000-4-2:2009	Electrostatic Discharge Test (ESD) / *Elektrostatik Boşalma Testi*	PASS/*GEÇTİ*
EN 61000-6-3:2007/A1:2011/AC:2012	Radiated Emission Test / *Yayılım Bozulması Testi*	PASS/*GEÇTİ*
EN 61000-6-3:2007/A1:2011/AC:2012	Conducted Emission Testi / *İletim Bozulması Testi*	PASS/*GEÇTİ*

EMC Deneyleri EMC Tests	**Deney Standardları** Test Standards **(TS / EN / IEC)**
1 - Elektrostatik Boşalma Bağışıklık Deneyi (Electrostatic Discharge Immunity Test)	**61000-4-2**
2 - Ani Yükselmelere Karşı Bağışıklık Deneyi (Surge Immunity Test)	**61000-4-5**
3 - RF Alanlar Tarafından Endüklenen,İletilen Bozulmalara Karşı Bağışıklık Deneyi (Immunity to Conducted Disturbances Induced by Radio Frequency Fields)	**61000-4-6**
4 - Işıyan, Radyo Frekans, Elektromanyetik Alan Bağışıklık Deneyi (Radiated, Radio Frequency, Electromagnetic Field Immunity Test)	**61000-4-3**
5 - Elektriksel Hızlı Geçici Rejim / Patlama Bağışıklık Deneyi (Electrical Fast Transient / Burst Immunity Test)	**61000-4-4**
6 - Gerilim Çukurları, Kısa Kesintiler ve Gerilim Değişmeleri Bağışıklık Deneyi (Voltage Dips, Short Interruptions and Voltage Variations Immunity Test)	**61000-4-11**

8. CONTROL UNIT SOCKET ENTRIES



9. TERMS OF WARRANTY

You can return the product with a petition containing the product invoice and the reason for the return within the period of seven (7) days from the date of delivery without being destroyed / spoiled before using the product you received (it can be used for trial purposes but its originality should not be impaired).

Open and check the packages that you think are damaged during shipment in front of the cargo authority you received. If there is any damage to the product, do not take delivery of the product by keeping a record with the cargo company. Please note that after the delivery of the product, you acknowledge that the cargo company has fully fulfilled its duties.

Returned products whose original packaging has been damaged, destroyed or otherwise cannot be accepted.

The customer is obliged to return the product to the condition it was delivered to, and to compensate for the loss of the commercial value of the product due to use.

If the manufacturer detects any opening, deterioration, breakage, destruction, rupture or use of the product and its packaging and if the product is not returned to the customer at the time it is delivered, the product will not return the product and will not refund the price.

After the product reaches the customer, the manufacturer's authorized services should be consulted.

In case of malfunction of the goods within the warranty period, the period spent in repair is added to the warranty period. The repair period of the goods is maximum 7 working days.

The manufacturer accepts no responsibility for the delivery of devices sent for service within 3 months from the date of arrival.

In case of failure of the goods during the warranty period due to both material and workmanship and assembly faults, the repair will be made without any charge under the labor cost, replacement part cost or any other name.

Malfunctions caused by the use of the product in contradiction with the instructions in the operating manual and user errors are not covered by the warranty.